

JEE Mains 2019 Chapter wise Question Bank

Probability – Questions

Q1

An urn contains 5 red and 2 green balls. A ball is drawn at random from the urn. If the drawn ball is green, then a red ball is added to the urn and if the drawn ball is red, then a green ball is added to the urn; the original ball is not returned to the urn. Now, a second ball is drawn at random from it. The probability that the second ball is red is:

- (1) $\frac{21}{49}$ (2) $\frac{27}{49}$
(3) $\frac{26}{49}$ (4) $\frac{32}{49}$

9 Jan Evening

Q2

An unbiased coin is tossed. If the outcome is a head then a pair of unbiased dice is rolled and the sum of the numbers obtained on them is noted. If the toss of the coin results in tail then a card from a well-shuffled pack of nine cards numbered 1, 2, 3, ..., 9 is randomly picked and the number on the card is noted. The probability that the noted number is either 7 or 8 is:

- (1) $\frac{13}{36}$ (2) $\frac{15}{72}$
(3) $\frac{19}{72}$ (4) $\frac{19}{36}$

10 Jan Morning

Q3

If the probability of hitting a target by a shooter, in any shot, is $\frac{1}{3}$, then the minimum number of independent shots at the target required by him so that the probability of hitting the target at least once is greater than $\frac{5}{6}$, is:

- (1) 3 (2) 6
(3) 5 (4) 4

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Q4

Two integers are selected at random from the set $\{1, 2, \dots, 11\}$. Given that the sum of selected numbers is even, the conditional probability that both the numbers are even is :

- (1) $\frac{7}{10}$ (2) $\frac{1}{2}$
(3) $\frac{2}{5}$ (4) $\frac{3}{5}$

11 Jan Morning

Q5

Let $S = \{1, 2, \dots, 20\}$. A subset B of S is said to be “nice”, if the sum of the elements of B is 203. Then the probability that a randomly chosen subset of S is “nice” is :

- (1) $\frac{7}{2^{20}}$ (2) $\frac{5}{2^{20}}$
(3) $\frac{4}{2^{20}}$ (4) $\frac{6}{2^{20}}$

11 Jan Evening

Q6

Probability

In a random experiment, a fair die is rolled until two fours are obtained in succession. The probability that the experiment will end in the fifth throw of the die is equal to :

- (1) $\frac{200}{6^5}$ (2) $\frac{150}{6^5}$
(3) $\frac{225}{6^5}$ (4) $\frac{175}{6^5}$

12 Jan Morning

Q7

In a game, a man wins ₹ 100 if he gets 5 or 6 on a throw of a fair die and loses ₹ 50 for getting any other number on the die. If he decides to throw the die either till he gets a five or a six or to a maximum of three throws, then his expected gain/loss (in rupees) is :

- (1) $\frac{400}{9}$ loss (2) 0
(3) $\frac{400}{3}$ gain (4) $\frac{400}{3}$ loss

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Q8

In a class of 60 students, 40 opted for NCC, 30 opted for NSS and 20 opted for both NCC and NSS. If one of these students is selected at random, then the probability that the student selected has opted neither for NCC nor for NSS is :

- (1) $\frac{1}{6}$ (2) $\frac{1}{3}$
(3) $\frac{2}{3}$ (4) $\frac{5}{6}$

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Q9

Let A and B be two non-null events such that $A \subset B$. Then, which of the following statements is always correct?

- (1) $P(A|B) = P(B) - P(A)$ (2) $P(A|B) \geq P(A)$
(3) $P(A|B) \leq P(A)$ (4) $P(A|B) = 1$

8 April Morning

Q10

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The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is :

- (1) 5 (2) 3 (3) 4 (4) 2

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Q11

Four persons can hit a target correctly with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is:

- (1) $\frac{25}{192}$ (2) $\frac{7}{32}$ (3) $\frac{1}{192}$ (4) $\frac{25}{32}$

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Q12

Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is:

- (1) $\frac{1}{11}$ (2) $\frac{1}{10}$ (3) $\frac{1}{12}$ (4) $\frac{1}{17}$

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Q13

Minimum number of times a fair coin must be tossed so that the probability of getting at least one head is more than 99% is :

- (1) 5 (2) 6 (3) 8 (4) 7

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Q14

If three of the six vertices of a regular hexagon are chosen at random, then the probability that the triangle formed with these chosen vertices is equilateral is :

- (1) $\frac{1}{10}$ (2) $\frac{1}{5}$ (3) $\frac{3}{10}$ (4) $\frac{3}{20}$

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Q15

Probability

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Let a random variable X have a binomial distribution with mean 8 and variance 4. If $P(X \leq 2) = \frac{k}{2^{16}}$, then k is equal to :

- (1) 17 (2) 121 (3) 1 (4) 137

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Q16

For an initial screening of an admission test, a candidate is given fifty problems to solve. If the probability that the candidate can solve any problem is $\frac{4}{5}$, then the probability that he is unable to solve less than two problems is :

- (1) $\frac{201}{5} \left(\frac{1}{5}\right)^{49}$ (2) $\frac{316}{25} \left(\frac{4}{5}\right)^{48}$
(3) $\frac{54}{5} \left(\frac{4}{5}\right)^{49}$ (4) $\frac{164}{25} \left(\frac{1}{5}\right)^{48}$

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Q17

A person throws two fair dice. He wins Rs. 15 for throwing a doublet (same numbers on the two dice), wins Rs. 12 when the throw results in the sum of 9, and loses Rs. 6 for any other outcome on the throw. Then the expected gain/loss (in Rs.) of the person is :

- (1) $\frac{1}{2}$ gain (2) $\frac{1}{4}$ loss
(3) $\frac{1}{2}$ loss (4) 2 gain

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Probability - Solutions

Q1

(4) Let G represents drawing a green ball and R represents drawing a red ball

So, the probability that second drawn ball is red

$$= P(G) \cdot P\left(\frac{R}{G}\right) + P(R)P\left(\frac{R}{R}\right)$$

$$= \frac{2}{7} \times \frac{6}{7} + \frac{5}{7} \times \frac{4}{7}$$

$$= \frac{12 + 20}{49}$$

$$= \frac{32}{49}$$

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Q2

$$(3) P(\text{Outcome is head}) = \frac{1}{2}$$

$$P(\text{Outcome is tail}) = \frac{1}{2}$$

$$P(7 \text{ or } 8 \text{ is the sum of two dice}) = \frac{6}{36} + \frac{5}{36}$$

$$= \frac{11}{36}$$

$$P(7 \text{ or } 8 \text{ is the number of card}) = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$\begin{aligned} \text{Required probability} &= \frac{1}{2} \times \frac{11}{36} + \frac{1}{2} \times \frac{2}{9} \\ &= \frac{1}{2} \left(\frac{11+8}{36} \right) = \frac{19}{72} \end{aligned}$$

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Q3

(3) Let the number of independent shots required to hit the target at least once be n , then

$$1 - \left(\frac{2}{3}\right)^n > \frac{5}{6}$$

$$\left(\frac{2}{3}\right)^n < \frac{1}{6}$$

Hence, the above inequality holds when least value of n is 5.

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Q4

(3) Probability of getting sum of selected two numbers is even

$$= P(E_1) = \frac{{}^5C_2 + {}^5C_2}{{}^{11}C_2}$$

Probability of getting sum is even and selected numbers are also even $P(E_2) = \frac{{}^5C_2}{{}^{11}C_2}$

$$\text{Hence, } P\left(\frac{E_2}{E_1}\right) = \frac{{}^5C_2}{{}^5C_2 + {}^5C_2} = \frac{10}{15+10} = \frac{2}{5}$$

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Q5

Probability

(2) Since total number of subsets of the set $S = 2^{20}$

Now, the sum of all number from 1 to 20 = $\frac{20 \times 21}{2} = 210$

Then, find the sets which has sum 7.

- (1) $\{7\}$
- (2) $\{1, 6\}$
- (3) $\{2, 5\}$
- (4) $\{3, 4\}$
- (5) $\{1, 2, 4\}$

Then, there is only 5 sets which has sum 203

Hence required probability = $\frac{5}{2^{20}}$

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Q6

(4) Since, the experiment will end in the fifth throw.

Hence, the possibilities are $4 * * 4 4$, $* 4 * 4 4$, $* * * 4 4$ (where $*$ is any number except 4)

$$\begin{aligned} \text{Required Probability} &= \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{1}{6}\right)\left(\frac{1}{6}\right) \\ &\quad + \left(\frac{5}{6}\right)\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)\left(\frac{1}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right)^2 \\ &= \frac{25 + 25 + 125}{6^5} = \frac{175}{6^5} \end{aligned}$$

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Q7

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(2) Probability of getting 5 or 6 = $P(E) = \frac{2}{6} = \frac{1}{3}$

Probability of not getting 5 or 6 = $P(E) = 1 - \frac{1}{3} = \frac{2}{3}$

E will consider as success.

Event	Success in I st attempt	Success in II nd attempt	Success in III rd attempt	No success in III rd attempt
Probability	$\frac{1}{3}$	$\frac{2}{3} \times \frac{1}{3}$	$\frac{2}{3} \times \frac{2}{3} \times \frac{1}{3}$	$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$
Gain/loss	100	50	0	-150

His expected gain/loss

$$\begin{aligned} &= \frac{1}{3} \times 100 + \frac{2}{9} \times 50 + \frac{8}{27} \times (-150) \\ &= \frac{900 + 300 - 1200}{27} = 0 \end{aligned}$$

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Q8

(1) P = Set of students who opted for NCC

Q = Set of Students who opted for NSS

$$n(P) = 40, n(Q) = 30, n(P \cap Q) = 20$$

$$n(P \cup Q) = n(P) + n(Q) - n(P \cap Q)$$

$$= 40 + 30 - 20$$

$$= 50$$

$\therefore$ Hence, required probability = $1 - \frac{50}{60}$

$$= \frac{1}{6}$$

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Q9

Probability

(2) $\because A \subset B$; so $A \cap B = A$

$$\text{Now, } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P\left(\frac{A}{B}\right) = \frac{P(A)}{P(B)}$$

$$\because P(B) \leq 1$$

$$\Rightarrow P\left(\frac{A}{B}\right) \geq P(A)$$

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Q10

(3) Let, p is probability for getting head and is probability for getting tail.

$$p = P(H) = \frac{1}{2}, q = 1 - p = \frac{1}{2}$$

$$P(x \geq 1) \geq \frac{9}{10} \Rightarrow 1 - P(x = 0) \geq \frac{9}{10}$$

$$1 - {}^nC_0 \left(\frac{1}{2}\right)^n \geq \frac{9}{10} \Rightarrow \frac{1}{2^n} \leq 1 - \frac{9}{10} \Rightarrow \frac{1}{2^n} \leq \frac{1}{10}$$

$$2^n \geq 10 \Rightarrow n \geq 4 \Rightarrow n_{\min} = 4$$

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Q11

(4) $P(\text{at least one hit the target}) = 1 - P(\text{none of them hit the target})$

$$= 1 - \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{8}\right)$$

$$= 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{7}{8} = 1 - \frac{7}{32} = \frac{25}{32}$$

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Q12

(1) Let, A = At least two girls

B = All girls

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{P(B)}{P(A)}$$

$$= \frac{\left(\frac{1}{4}\right)^4}{1 - {}^4C_0 \left(\frac{1}{2}\right)^4 - {}^4C_1 \left(\frac{1}{2}\right)^4} = \frac{1}{16 - 1 - 4} = \frac{1}{11}$$

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Q13

(4) According to the question,

$$1 - \left(\frac{1}{2}\right)^n > \frac{99}{100} \Rightarrow \left(\frac{1}{2}\right)^n < \frac{1}{100} \Rightarrow n \geq 7$$

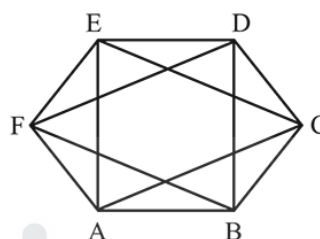
Hence, minimum value is 7.

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Q14

(1) Total no. of triangles = 6C_3

Favorable no. of triangle i.e., equilateral triangles ($\triangle AEC$ and $\triangle BDF$) = 2.



$$\text{Hence, required probability} = \frac{2}{{}^6C_3} = \frac{2}{10}$$

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Q15

(4) Given mean $\mu = 8$ and variance $\sigma^2 = 4$

$$\Rightarrow \mu = np = 8 \text{ and } \sigma^2 = npq = 4.$$

$$p + q = 1 \Rightarrow q = \frac{1}{2}, p = \frac{1}{2} \text{ and } n = 16$$

$$\therefore P(X \leq 2) = \frac{k}{2^{16}}$$

$$\therefore {}^{16}C_0 \left(\frac{1}{2}\right)^{16} + {}^{16}C_1 \left(\frac{1}{2}\right)^{16} + {}^{16}C_2 \left(\frac{1}{2}\right)^{16} = \frac{k}{2^{16}}$$

$$\Rightarrow k = (1 + 16 + 120) = 137$$

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Q16

- (3) Let p is the probability that candidate can solve a problem and q is the probability that candidate can not solve a problem.

$$p = \frac{4}{5} \text{ and } q = \frac{1}{5} \quad (\because p + q = 1)$$

Probability of solving either 50 or 49 problem by the candidate

$$= {}^{50}C_{50} \cdot p^{50} \cdot q^0 + {}^{50}C_{49} \cdot p^{49} \cdot q^1 = p^{49} [p + 50q]$$

$$= \left(\frac{4}{5}\right)^{49} \cdot \left(\frac{4}{5} + \frac{50}{5}\right) = \frac{54}{5} \cdot \left(\frac{4}{5}\right)^{49}$$

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Q17

- (3) Let X be the random variable which denotes the Rs gained by the person.

Total cases = $6 \times 6 = 36$.

Favorable cases for the person on winning ₹ 15 are (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) i.e., 6 cases.

$$\therefore P(X=15) = \frac{6}{36} = \frac{1}{6}$$

Favorable cases for the person on winning ₹ 12 are (6, 3), (5, 4), (4, 5), (3, 6) i.e., 4.

$$\therefore P(X=12) = \frac{4}{36} = \frac{1}{9}$$

Remaining cases = $36 - 6 - 4 = 26$

$$\therefore P(X=-6) = \frac{26}{36} = \frac{13}{18}$$

X	15	12	-6
$P(X)$	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{13}{18}$
$X \cdot P(X)$	$\frac{5}{2}$	$\frac{4}{3}$	$-\frac{13}{3}$

$$\text{Hence, } E(X) = \sum X \cdot P(X) = \frac{5}{2} + \frac{4}{3} - \frac{13}{3} = -\frac{1}{2}$$

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